

MGTA 457: Business Intelligence Systems

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This is a hands-on, project-based course designed to transform high-volume data from disparate systems into real-time, actionable business insights. We will utilize the Databricks Data Intelligence Platform for scalable storage and processing, alongside Power BI for visual analytics. The principles of modern BI architecture, moving from lakehouse to dashboard, are the core of this course.

Contact Information

- **Instructor:** Vikram Jambulapati
- **Email:** v2jambulapati@ucsd.edu
- **Office Hours:** By appointment (Zoom)
- **TA:** TBD

Course Objectives

At the close of this course, you will be able to:

- Articulate the value of cloud-native BI systems.
- Manage data lifecycles using Databricks (Loading, Scaling, and Querying).
- Employ infographic design principles to convey information rapidly.
- Recognize when design or data modeling may lead to poor interpretation.
- Combine, cleanse, and model data feeds for production-grade dashboards.

Grading Breakdown

Component	%
Presentation	40
Databricks/Data Prep	25
Final Document	15
Design Doc	10
Quizzes	10

Required Materials

- **Hardware:** Windows 10+ laptop recommended (Power BI Desktop is Windows-only; Mac users can use the Power BI web service or a Windows VM).
- **Software:** Power BI Desktop (free) and Databricks (Free Edition trial workspaces set up in class).
- **AI Policy:** Use of Generative AI (ChatGPT, Copilot, Databricks Assistant, etc.) is expected for SQL optimization and dashboard calculation logic.

Schedule of Activities

Week	Topic	Assignments Due (Before Class)
Week 6	Intro to BI & The Data Platform	Course Overview; Databricks Account Setup
Week 7	Databricks Foundations: SQL & ELT	Lab: Databricks Overview
Week 8	Visual Principles & Power BI Connection	Dashboard Design Document (1 per team)
Week 9	Advanced ETL: Power Query & Databricks	Read: Shape and Combine Data in Power BI Desktop
Week 10	Student Team Presentations	Final Project Documentation & Dashboard Files
Finals	Student Team Presentations	Peer and Team Reviews

Team Project Deliverables

- **Dashboard Design Document:** 2-5 pages with sketches and intended user flow.
- **Databricks Lakehouse:** A curated set of Unity Catalog tables containing your cleaned, joined datasets.
- **Power Query / SQL Workflow:** Demonstration of the transformation layer (either via SQL in Databricks or Power Query in Power BI).
- **Final Presentation:** 10–15 minute story using at least 3 dashboards (Power BI reports) and 6+ visuals.

Policies & Integrity

Punctuality: Class starts on time; attendance counts toward participation.

Academic Integrity: While AI use is encouraged, all logic and visualizations must be explainable by the team during the presentation.